

Please enter a MongoDB connection string (Default: mongodb://localhost/):

Current Mongosh Log ID: 69f4b1d95774e1df0a3682d0

Connecting to: `mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2`

Using MongoDB: 7.0.31

Using Mongosh: 2.8.2

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting
2026-05-01T19:17:33.504+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

```
test> use library_db
switched to db library_db
library_db> db.books.drop();
| db.members.drop();
| db.loans.drop();
true
library_db> db.books.insertMany([
  { "_id": "b1",
    "title": "The Hidden Code", "author": "Sara
Lee", "genre": "Mystery", "copies": 5},

  { "_id": "b2",
    "title": "Skyward Journey", "author": "Tom
Clark", "genre": "Sci-Fi", "copies": 3},

  { "_id": "b3",
    "title": "Ancient Tales", "author": "Emma
Stone", "genre": "History", "copies": 7},

  { "_id": "b4",
    "title": "Code Basics", "author": "John
Doe", "genre": "Technology", "copies": 4},
```

```

library_db> db.books.insertMany([
  { "_id": "b1", "title": "The Hidden Code", "author": "Sara Lee", "genre": "Mystery", "copies": 5},
  { "_id": "b2", "title": "Skyward Journey", "author": "Tom Clark", "genre": "Sci-Fi", "copies": 3},
  { "_id": "b3", "title": "Ancient Tales", "author": "Emma Stone", "genre": "History", "copies": 7},
  { "_id": "b4", "title": "Code Basics", "author": "John Doe", "genre": "Technology", "copies": 4},
  { "_id": "b5", "title": "Poems of Light", "author": "Lily Hart", "genre": "Poetry", "copies": 6}
]);

{
  acknowledged: true,
  insertedIds: { '0': 'b1', '1': 'b2', '2': 'b3', '3': 'b4', '4': 'b5' }
}

library_db> db.members.insertMany([
  { "_id": "m1", "name": "Alex Green", "email": "alex.green@lib.com", "join_date": ISODate("2022-03-10T00:00:00Z")},
  { "_id": "m2", "name": "Maria Lopez", "email": "maria.lopez@lib.com", "join_date": ISODate("2022-06-15T00:00:00Z")},
  { "_id": "m3", "name": "Sam Patel", "email": "sam.patel@lib.com", "join_date": ISODate("2023-01-20T00:00:00Z")}
]);

{
  acknowledged: true,
  insertedIds: { '0': 'm1', '1': 'm2', '2': 'm3' }
}

library_db> db.loans.insertMany([
  { "_id": "l1", "member_id": "m1", "book_id": "b2", "loan_date": ISODate("2023-02-01T10:00:00Z"), "return_date": ISODate("2023-02-15T00:00:00Z")},
  { "_id": "l2", "member_id": "m2", "book_id": "b1", "loan_date": ISODate("2023-02-05T14:30:00Z"), "return_date": null},
  { "_id": "l3", "member_id": "m3", "book_id": "b4", "loan_date": ISODate("2023-02-10T09:15:00Z"), "return_date": null},
  { "_id": "l4", "member_id": "m1", "book_id": "b3", "loan_date": ISODate("2023-02-12T11:00:00Z"), "return_date": ISODate("2023-02-20T00:00:00Z")},
  { "_id": "l5", "member_id": "m2", "book_id": "b5", "loan_date": ISODate("2023-02-15T16:00:00Z"), "return_date": null}
]);

{
  acknowledged: true,
  insertedIds: { '0': 'l1', '1': 'l2', '2': 'l3', '3': 'l4', '4': 'l5' }
}

library_db> db.books.insertOne({
  "_id": "b6",
  "title": "New Horizons",
  "author": "Jane Smith",
  "genre": "Adventure",
  "copies": 2
});

```

```

| });
{ acknowledged: true, insertedId: 'b6' }
library_db> db.books.find({ _id: "b6" }).pretty();
[
  {
    _id: 'b6',
    title: 'New Horizons',
    author: 'Jane Smith',
    genre: 'Adventure',
    copies: 2
  }
]
library_db> db.members.find({
|   join_date: { $gt: ISODate("2023-01-01T00:00:00Z") }
| }).pretty();
[
  {
    _id: 'm3',
    name: 'Sam Patel',
    email: 'sam.patel@lib.com',
    join_date: ISODate('2023-01-20T00:00:00.000Z')
  }
]
library_db> db.members.updateOne(
|   { name: "Alex Green" },
|   { $set: { email: "alex.newemail@lib.com" } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library_db> db.members.find({ name: "Alex Green" }).pretty();
[
  {
    _id: 'm1',
    name: 'Alex Green',
    email: 'alex.newemail@lib.com',
    join_date: ISODate('2022-03-10T00:00:00.000Z')
  }
]

```

```

library_db> db.loans.deleteOne({
  member_id: "m2",
  book_id: "b1"
});
{ acknowledged: true, deletedCount: 1 }
library_db> db.books.createIndex({ title: 1 });
title_1
library_db> db.books.find({ title: "Code Basics" });
[
  {
    _id: 'b4',
    title: 'Code Basics',
    author: 'John Doe',
    genre: 'Technology',
    copies: 4
  }
]
library_db> db.members.createIndex({ name: 1 });
name_1
library_db> db.members.find({ name: "Alex Green" });
[
  {
    _id: 'm1',
    name: 'Alex Green',
    email: 'alex.newemail@lib.com',
    join_date: ISODate('2022-03-10T00:00:00.000Z')
  }
]
library_db> db.loans.createIndex({ member_id: 1, book_id: 1 });
member_id_1_book_id_1
library_db> db.books.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { title: 1 }, name: 'title_1' }
]
library_db> db.members.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1 }, name: 'name_1' }
]

```

```

library_db> db.books.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { title: 1 }, name: 'title_1' }
]
library_db> db.members.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1 }, name: 'name_1' }
]
library_db> db.loans.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { member_id: 1, book_id: 1 },
    name: 'member_id_1_book_id_1'
  }
]
library_db> db.loans.aggregate([
  {
    $group: {
      _id: "$book_id",
      borrow_count: { $sum: 1 }
    }
  },
  {
    $lookup: {
      from: "books",
      localField: "_id",
      foreignField: "_id",
      as: "book_details"
    }
  },
  {
    $unwind: "$book_details"
  },
  {
    $project: {
      _id: 0,
      title: "$book_details.title",

```

```

library_db> db.loans.aggregate([
  {
    $group: {
      _id: "$member_id",
      total_loans: { $sum: 1 }
    }
  },
  {
    $group: {
      _id: null,
      average_loans_per_member: { $avg: "$total_loans" }
    }
  },
  {
    $project: {
      _id: 0,
      average_loans_per_member: 1
    }
  }
]);
[ { average_loans_per_member: 1.3333333333333333 } ]
library_db> db.loans.aggregate([
  {
    $lookup: {
      from: "books",
      localField: "book_id",
      foreignField: "_id",
      as: "book_details"
    }
  },
  {
    $unwind: "$book_details"
  },
  {
    $group: {
      _id: "$book_details.genre",
      total_loans: { $sum: 1 }
    }
  }
]);

```

```

    $project: {
      _id: 0,
      average_loans_per_member: 1
    }
  }
  ]});
[ { average_loans_per_member: 1.3333333333333333 } ]
library_db> db.loans.aggregate([
  {
    $lookup: {
      from: "books",
      localField: "book_id",
      foreignField: "_id",
      as: "book_details"
    }
  },
  {
    $unwind: "$book_details"
  },
  {
    $group: {
      _id: "$book_details.genre",
      total_loans: { $sum: 1 }
    }
  },
  {
    $sort: { total_loans: -1 }
  },
  {
    $limit: 1
  },
  {
    $project: {
      _id: 0,
      genre: "$_id",
      total_loans: 1
    }
  }
  ]});
[ { total_loans: 1, genre: 'History' } ]
library_db> |

```